

Evolutionary Biology

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

About eLife's process

Reviewed preprint posted

July 3, 2023 (this version)

Sent for peer review

May 3, 2023

Posted to bioRxiv

April 14, 2023

Repeatability of adaptation in sunflowers: genomic regions harbouring inversions also drive adaptation in species lacking an inversion

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Abstract

Local adaptation commonly involves alleles of large effect, which experience fitness advantages when in positive linkage disequilibrium (LD). Because segregating inversions suppress recombination and facilitate the maintenance of LD between locally adapted loci, they are also commonly found to be associated with adaptive divergence. However, it is unclear what fraction of an adaptive response can be attributed to inversions and alleles of large effect, and whether the loci within an inversion could still drive adaptation in the absence of its recombination-suppressing effect. Here, we use genome-wide association studies to explore patterns of local adaptation in three species of sunflower: *Helianthus annuus*, *H. argophyllus*, and *H. petiolaris*, which each harbour a large number of species-specific inversions. We find evidence of significant genome-wide repeatability in signatures of association to phenotypes and environments, which are particularly enriched within regions of the genome harbouring an inversion in one species. This shows that while inversions may facilitate local adaptation, at least some of the loci involved can still make substantial contributions without the benefit of recombination suppression. While a large number of genomic regions show evidence of repeated adaptation, most of the strongest signatures of association still tend to be species-specific, indicating substantial genotypic redundancy for local adaptation in these species.

eLife assessment

This is a **valuable** comparative study of local adaptation using gene-by-environment and gene-by-phenotype correlations. The analyses seemed still **incomplete**, as the biological take-home messages were obscured by the statistical approaches used, and it remains unclear how to best interpret the level of genome-wide convergence and in inversions. The repeatability of local adaptation across species, and the role of inversions in local adaptation, are questions of considerable empirical interest.

Introduction

The genetic basis of local adaptation is sometimes highly repeatable, with examples of large effect genes driving responses in multiple species, such as *FT* affecting flowering time in numerous plants (Izawa 2007; Auge et al. 2019) or *Mc1r* driving colour polymorphisms in vertebrates (Manceau et al. 2010; Rosenblum et al. 2014). Local adaptation can also be repeated for polygenic traits, with significant patterns of similar association found across many loci for comparisons of conifers (Yeaman et al. 2016), maize and its wild relative teosinte (Tittes et al. 2021; Wang et al. 2021), and *Brassicaceae* (Bohutínská et al. 2021), to name a few. While we have a growing number of examples of repeatability in the basis of adaptation, it is also interesting to know if species use different genes to adapt to the same selection pressure. Genotypic redundancy – the potential for many genotypes to yield a given phenotype – is one critical factor affecting the repeatability of adaptation, as high redundancy would be expected to result in lower repeatability (Yeaman et al. 2018). Another critical factor affecting repeatability is shared standing variation, whether present due to introgression or incomplete lineage sorting. In either case, variants shared among lineages are much more likely to contribute to a repeated response than new mutations (MacPherson and Nuismer 2018; Ralph and Coop 2015). Consistent with this, Bohutínská and colleagues (2021) found that repeatability was negatively related to phylogenetic distance. While we are now accumulating more studies about the repeatability of adaptation, we still have very few examples and much remains unknown about the relative importance of these factors (Yeaman 2022).

Inversions have been implicated in local adaptation in many species (Wellenreuther and Bernatchez 2018), likely due to their effect to suppress recombination between any causal loci within their bounds (Rieseberg 2001; Noor et al. 2001; Kirkpatrick and Barton 2006). Despite mounting evidence of their importance in adaptation, it is unclear how inversions may covary with repeatability of adaptation among species. If most alleles have small effects relative to migration rate and can only contribute to local adaptation via the benefit of the recombination-suppressing effect of an inversion, then we would expect little repeatability at the site of an inversion – other species lacking the inversion would not tend to use that same region for adaptation. On the other hand, if some loci are particularly important for local adaptation and regularly yield mutations of large effect, with these patterns being conserved among species, repeatability within regions harbouring inversions may be substantial.

Here, we explore the repeatability of local adaptation in three species of sunflowers, *Helianthus annuus*, *Helianthus argophyllus* and *Helianthus petiolaris* (Figure 1), which harbour large regions of suppressed recombination (“haploblocks”), most of which are inversions, and are often associated with adaptive traits (Todesco et al. 2020). *Helianthus annuus*, the common sunflower, is the closest wild relative of cultivated sunflower, which was domesticated from it around 4,000 years ago (Blackman et al. 2011). Populations of *H. annuus* are distributed throughout the central and western USA and generally found on mesic soils, but can grow in a variety of disturbed or extreme habitats, such as semi-desertic or frequently flooded areas, as well as salt marshes. *Helianthus petiolaris*, the prairie sunflower, prefers sandier soils, and ecotypes of this species are adapted to sand sheets and sand dunes (Ostevik et al. 2016). Here we include samples from two subspecies: *H. petiolaris* ssp. *petiolaris*, which is commonly found in the southern Great Plains, and *H. petiolaris* ssp. *fallax*, which is limited to more arid regions in Colorado, Utah, New Mexico and Arizona (Heiser et al. 1969). *Helianthus petiolaris* and *H. annuus* have broad and overlapping distributions throughout the central and western United States and appear to have adapted to similar changes in temperature, moisture and photoperiod regimes. There is also evidence indicating *H. annuus* and *H. petiolaris* have likely been exchanging genes during much of

their history of divergence (Strasburg and Rieseberg 2008), although partially isolated by strong pre- and post-zygotic barriers (Sambatti et al. 2012). The third species, *Helianthus argophyllus*, the silverleaf sunflower, is found exclusively in the southeast coast of Texas and includes both an early flowering ecotype on the coastal barrier islands and a late flowering ecotype inland (Moyers and Rieseberg 2016; Todesco et al. 2020). *H. argophyllus* is thought to have undergone cycles of sympatry and allopatry with *H. annuus* and *H. petiolaris* over time, but currently only overlaps with *H. annuus*, with this overlap likely being a recent event (Heiser, 1951).

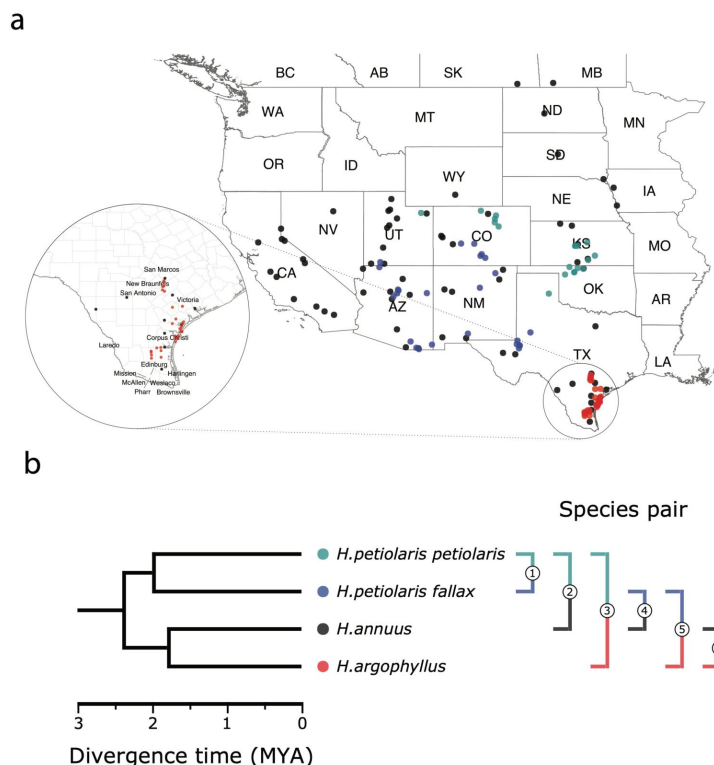


Figure 1.

Sampling sites and phylogenetic relationship among surveyed species.

(a) Sampling locations of wild sunflower populations studied in the present study, and (b) phylogenetic relationship of the four (sub-)species. Numbered brackets represent the six pairwise comparisons performed in this study.

Using broad sampling across the ranges of these species (Figure 1) and sequencing data first published by Todesco et al. (2020), we study the genomic basis of local adaptation by conducting genome-wide scans for associations with climatic and soil environmental variables and phenotypes measured in a common garden. An inherent problem in studying the basis of local adaptation is accounting for the covariance between genome-wide population genetic structure and environment. When the selection pressure driving local adaptation tends to parallel a main axis of demographic expansion or isolation by distance, neutral alleles will have spatial patterns resembling those of causal alleles. Methods that do not correct for population structure have large numbers of false positives because allele frequencies at neutral loci tend to correlate with the environment more than expected by chance (Lotterhos and Whitlock 2014). By contrast, methods that use structure correction tend to suffer from false negatives because the causal loci have similar patterns of allele frequency variation as the genomic background, and so their statistical signatures are decreased (DeRaad et al. 2021; Booker et al. 2023b). When a non-corrected approach is applied to multiple species, the false positive problem can be mitigated when testing for loci that contribute to repeated adaptation, as the same gene should not tend to be a false positive in multiple species more often than expected by chance (Yeaman et al. 2016). Here we use a conventional GWAS approach with structure correction to study the basis of phenotypes, which vary both within and among populations, and use an uncorrected approach to test association to environment, which only varies among populations (as all

individuals within the same population have the same value of environmental variable). For the environmental association portions of the study, we also explore the effect of structure correction in some analyses, to contrast the false negative vs. false positive problem inherent in each approach. We apply these approaches individually to *H. annuus*, *H. argophyllus*, and the two *H. petiolaris* subspecies, and make pairwise and higher-order comparisons among combinations of these four lineages to study repeatability.

Our overall aim is to characterize the extent of repeatability of local adaptation for different traits and environments, and explore the importance of inversions and other factors that may drive differences in repeatability. This analysis builds upon the work of Todesco et al. (2020), which focused on the evolution of haploblocks within each species, but did not systematically study patterns of repeatability in signatures of association among species. We begin by using the strength of correlations between environment and phenotypes measured in a common garden to identify which environmental variables are likely driving local adaptation within each species, and use an index to quantify the relative similarity in these patterns among pairs of species. We then conduct Genome-Wide Association tests to identify regions most strongly associated with phenotypes and environments within each species. To assess the similarity in these association statistics among species, we use methods similar to Yeaman et al. (2016) to identify regions of the genome that exhibit greater similarity in signatures of association among pairs of lineages than expected by chance, and explore how the number and total size of these regions covaries with the index of similarity in environmental importance. We also explore whether these regions of repeated association tend to covary with patterns of shared standing variation and test if they are enriched within previously identified haploblocks, to explore the role of inversions in adaptation. We then identify candidate genes that show particularly strong signatures of repeated association across multiple lineages. Finally, based on the number of signatures of association that are shared vs. non-shared among lineages, we estimate the number of regions genome-wide that could potentially contribute to local adaptation for each variable and phenotype, which is related to the genotypic redundancy. Taken together, our results show that local adaptation in sunflowers tends to involve both strong repeatability at a small number of genes, often associated with inversions, coupled with high redundancy and non-repeated responses across a much larger number of loci.

Results

Local adaptation at the phenotypic level.

The four taxa vary considerably in the breadth of environmental variation spanned by their respective ranges, with *H. annuus* spanning the widest niche and *H. argophyllus* spanning the narrowest (Figure 2A). We observe strong correlations for many combinations of phenotypes and environmental variables (Figure S1), suggesting that local adaptation is quite pronounced for some traits. To quantify the similarity in patterns of local adaptation among pairs of species, for each environmental variable we calculate an index we refer to as the Similarity in Phenotype-Environment Correlation (SIPEC), which is maximized when both species have strong correlations with the environmental variable across many phenotypes (see Methods). To prevent heavily weighting the contribution of multiple highly correlated phenotypes, we fit a principal components analysis to the phenotypes and used the resulting axes to calculate SIPEC. We find the largest values of SIPEC for temperature variables and the smallest values for soil types, and generally find higher values for comparisons between the *petiolaris* subspecies (Figure S2).

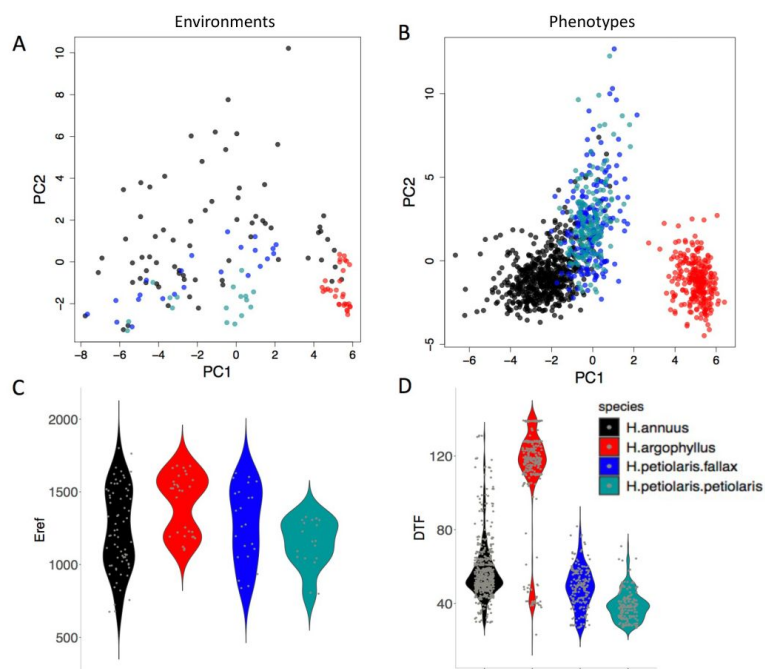


Figure 2.

The range of environmental and phenotypic variation in the studied species.

Variation in environment (a) and phenotype (b) for the studied species, along the two largest axes of a Principle component analysis (PCA). Violin plots show two examples of variation in environment (Hargreaves reference evapotranspiration index; Eref) and phenotype (Days to Flower; DTF) within and among the taxa (c,d).

Genome-wide analysis of repeated local adaptation

To search for regions of the genome driving repeated adaptation in multiple taxa, we first identified windows of the genome within each species that showed strong signatures of association to either phenotype (GWAS) or environment (GEA), referred to as “top candidate” windows. To identify Windows of Repeated Association (WRAs) between pairs of species, we then assessed whether top candidate windows identified in one focal taxon also tended to be enriched for strong signatures of association to the same environment or phenotype in each of the other taxa (using the null-W test, which tends to be more sensitive than just finding the overlap between the top candidate windows; See Methods). As recombination rate can affect the sensitivity of these kinds of window-based genome scans (Booker et al. 2020), the identification of WRAs was conducted after binning windows by recombination rate, although in many cases we did not observe substantial differences in the null distributions among these bins (e.g., Figure S3). This analysis revealed many windows with signatures of repeated association, with the strongest repeated signature found for Number of Frost-Free Days (NFFD; Figure 3A; S4), but also showed that many of the windows with the strongest association in one species did not have strong signatures in other species (Figure 3B).

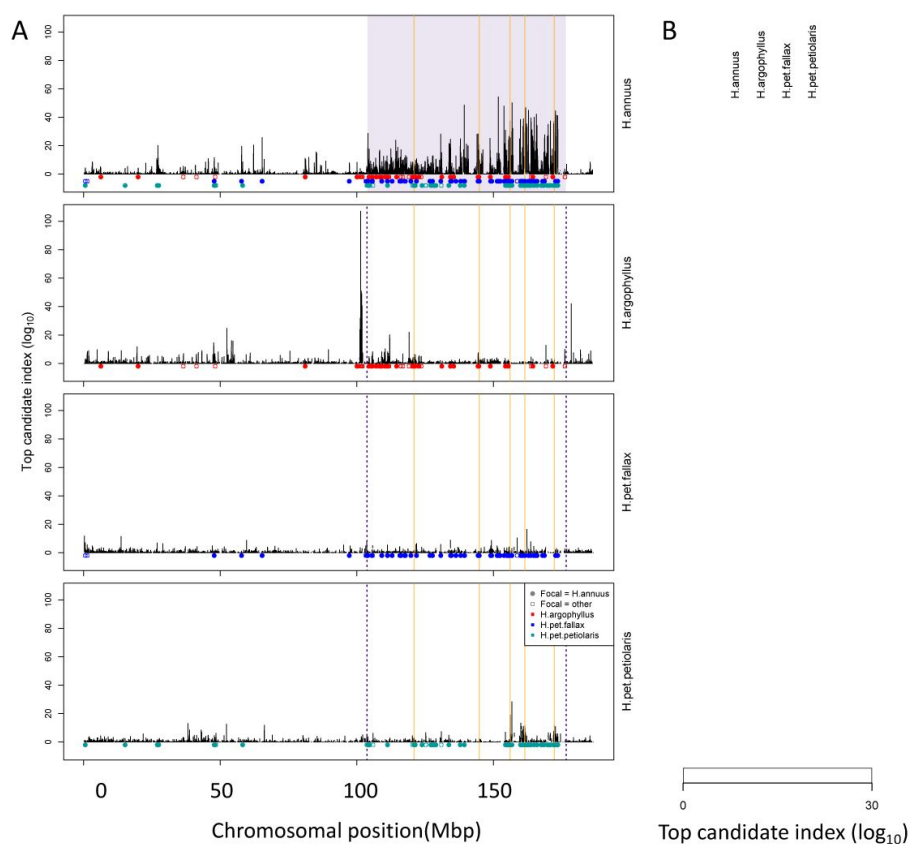


Figure 3.

Signatures of association for Number of Frost-Free Days (NFFD) in the four taxa on chromosome 15 (A) and genome-wide (B). Panel A shows Windows of Repeated Association (WRAs; coloured bars) for comparisons between the focal species, *H. annuus*, and each of the other 3 taxa, with the haploblocks in *H. annuus* shaded in violet and the regions with significant PicMin hits as vertical orange lines. Panel B shows the value of the top candidate index for each of the 1000 windows with the strongest signatures of association in at least one species (~top 2% of genome-wide windows). Rows are ordered using hierarchical clustering to group windows with similar patterns across multiple species, illustrating the extent of overlap/non-overlap in the windows with strongest signatures in each species (*i.e.*

position in the figure does not reflect chromosomal position).

Under the null hypothesis that all regions of the genome evolve neutrally due to drift and independently in each species, ~5% of the top candidate windows identified in one species would be expected to fall into the tail of the null distribution of another species (*i.e.* be classified as WRAs). While we find significantly more WRAs than the 5% expected by chance, ranging from 6.3% to 44.1% (mean = 14.5%; Figure S4), multiple factors can violate the assumptions of this test and increase the proportion of windows classified as WRAs. Most importantly, similarity in signatures of association can be driven by shared ancestral variation or ongoing introgression, and this must always be considered as an alternative explanation. It is clear that most regions of the genome harbour some shared standing variation between all 6 pairwise comparisons of the taxa, whether due to segregating ancestral variation or introgression, based on an index quantifying the proportion of shared to non-shared SNPs in each window (Figure S5). However, we do not find consistent differences in the amount of shared standing variation within windows when comparing WRAs with the rest of the genome (Figure S5). Thus, while introgression may make subtle contributions to WRAs, a lack of increased shared standing variation in the significant windows suggests that it is not a primary driver of these patterns.

A second source of inflation in the number of WRAs is due to hitchhiking: if a given locus is a true positive driving adaptation, LD with other windows in tight physical linkage will result in spurious correlations causing them to also be classified as WRAs increasing the genome-wide proportion above 5%, and this effect will be particularly exaggerated in haploblocks. To control for this, we binned neighbouring WRAs together that exhibited high LD (>95th percentile) over short spans of the genome (<1 cM) in either species, yielding regions we

refer to as Clusters of Repeated Association (CRAs). While this clustering method should provide a partial control for the effect of hitchhiking, we note that the number of CRAs cannot be taken as a reliable estimate of the number of independent targets of natural selection. Instead, we treat the number and size of these CRAs as proxies for the relative genome-wide similarity in association between each pair of species, and conduct downstream analyses to test hypotheses about the factors driving local adaptation.

CRAs varied in number and size across the 6 pairwise-species comparisons, with particularly large CRAs identified at the sites of known haploblocks (Todesco et al. 2020). For CRAs identified for phenotypic associations, their extent ranged from 168 clusters spanning 48,524,903 bp (covering only 1.6% of the genome) in *H. argophyllus-H. pet. petiolaris* to a maximum of 1,667 clusters spanning 426,409,647 bp (14.2 % of the genome) in *H. pet. petiolaris-H. pet. fallax* (Figure S6). CRAs identified for environmental variables tended to be more numerous and cover a larger region of the genome, varying from 1154 clusters comprising 15% of the total genome for *H. argophyllus-H. pet. fallax* up to 2,260 distinct clusters covering 29% of the genome between *H. petiolaris* subspecies (Figure S7). The large extent of the genome covered by CRAs is mainly driven by their occurrence within many different haploblocks, which tend to be unique to each species and cover substantial portions their genomes (Todesco et al. 2020). Genes within CRAs tend to be enriched for a large number of GO terms spanning many different categories (Figure S8, S9). When we use an approach that corrects for population structure in the genotype-environment association tests (Baypass), we find substantially reduced numbers of CRAs (Figure S10). This reduction is likely due to reduced power to detect true positive causal loci when population structure covaries with environment, as true positives do not “stand out” relative to the genomic background (Booker et al. 2023b). While single-species association tests without correction for population structure typically yield large numbers of false positives, the null-W test comparing association results across species accounts for the chance that the same region is a false positive in both species (Yeaman et al. 2016).

Repeated association in the genome reflects patterns at the phenotypic level

If natural selection is driving repeated patterns of local adaptation in the genomes of two species, we should see greater similarity for associations with environmental variables that are also strongly associated with phenotypic variation in both species. Consistent with this prediction, we found that environmental variables with a high SIPEC index (i.e. high correlation with phenotypes; Figure S2) also tended to have a larger number of CRAs in all comparisons (Figure 4), with higher repeatability at both the phenotype and genome level for temperature-related variables and much lower repeatability for soil-related variables. Unfortunately, given the non-independence of environmental variables it is not possible to conduct a formal significance test of these patterns. In all cases, the number of CRAs exhibited a weak negative relationship with the index of standing variation, with environmental variables that had the greatest number of CRAs having the lowest relative amounts of shared standing variation (Figure S11). This suggests that the observed similarity in patterns of association among species is not being strongly driven by incomplete lineage sorting or introgression, as we would expect a positive relationship between the index of shared standing variation and the number of CRAs. We found similar but weaker patterns in the association between SIPEC and size of CRAs (Figure S12), which may reflect comparatively greater contribution of haploblocks driving patterns with size rather than number (i.e. if a particularly large haploblock is a CRA, then that variable will have a particularly large value relative to its actual importance for local adaptation).

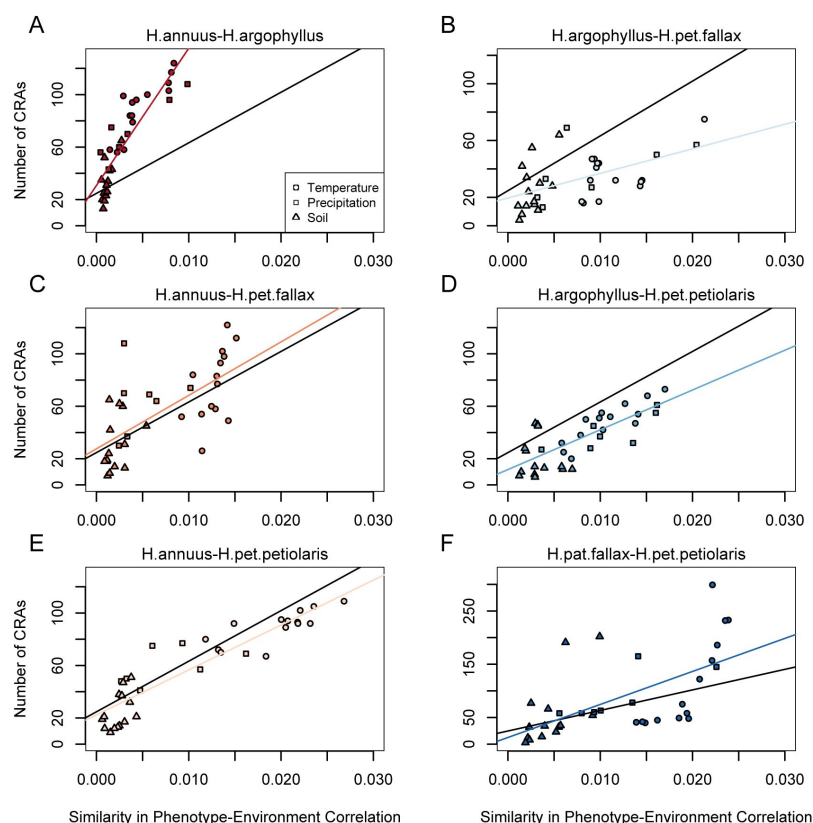


Figure 4.

Relationship between mean Similarity in Phenotype-Environment Correlation (SIPEC) and number of Clusters of Repeated Association (CRAs). Each panel includes both a linear model fit to the data within the panel (coloured lines), and a linear model fit to all data simultaneously (black lines) for comparison.

Overlap of signatures of repeated association with low recombination haploblocks

Inversions can facilitate local adaptation by suppressing recombination between locally adapted alleles within the inverted vs. ancestral haplotypes (Kirkpatrick and Barton 2006). If a given species exhibits a signature of association to environment at a segregating inversion, it is interesting to test whether similar signatures of association are found in other species that lack a segregating inversion at the same region of the genome. This would suggest that particularly strong selection is acting on this region, as signatures of association can still evolve even without the recombination suppressing effect of the inversion. While not all of the low-recombination haploblocks identified by Todesco et al. (2020) have been validated as inversions, for simplicity we treat each haploblock as representative of a segregating inversion. The majority of these haploblocks are present as segregating variation in only one of the three species, but when they occur in *H. petiolaris*, they tend to be found as segregating variation within both subspecies. Thus, if we find significant enrichment of the regions of repeated association (WRAs and CRAs) within haploblocks, this suggests that both the species with a segregating inversion and the species lacking an inversion are using this region of the genome to drive local adaptation. As a first test of this relationship, we assessed two-way contingency tables for whether top candidate windows within a species were also significant windows of repeated association (WRA/non-WRA) and whether they fell within haploblocks (yes/no). This approach controls for the potential enrichment of top candidate windows within haploblocks relative to the rest of the genome and asks if the rate of WRAs within haploblocks is higher. For most phenotypes and environments, we found that the proportion of WRAs in haploblocks was much higher than for non-WRAs (Figure 5). Across all species comparisons and environmental/phenotypic variables, a total of 310 contrasts showed significant enrichment for WRAs in haploblocks, compared to only 23 contrasts showing a significantly higher proportion of non-WRAs in haploblocks. In *H. argophyllus*,

while non-WRA top candidates tended to fall within haploblocks less commonly than in the other taxa, WRAs tended to be very strongly enriched within haploblocks (Figure 5B). As a follow-up, we also tested whether CRAs were enriched within haploblocks (without controlling for whether top candidate windows also tended to be enriched), also finding strongly significant enrichment of CRAs within haploblocks for many traits and environments (Figure S6C & D). Broad patterns revealed by these analyses are similar, as both show that regions with haploblocks tend to be enriched for signatures of association to environment in species lacking the haploblock. It is noteworthy that few significant signatures of enrichment in haploblocks were found for *H. petiolaris petiolaris* vs. *H. petiolaris fallax* (Figure S6C & D), perhaps because extensive introgression across non-locally adapted regions of the genome obscures true signal.

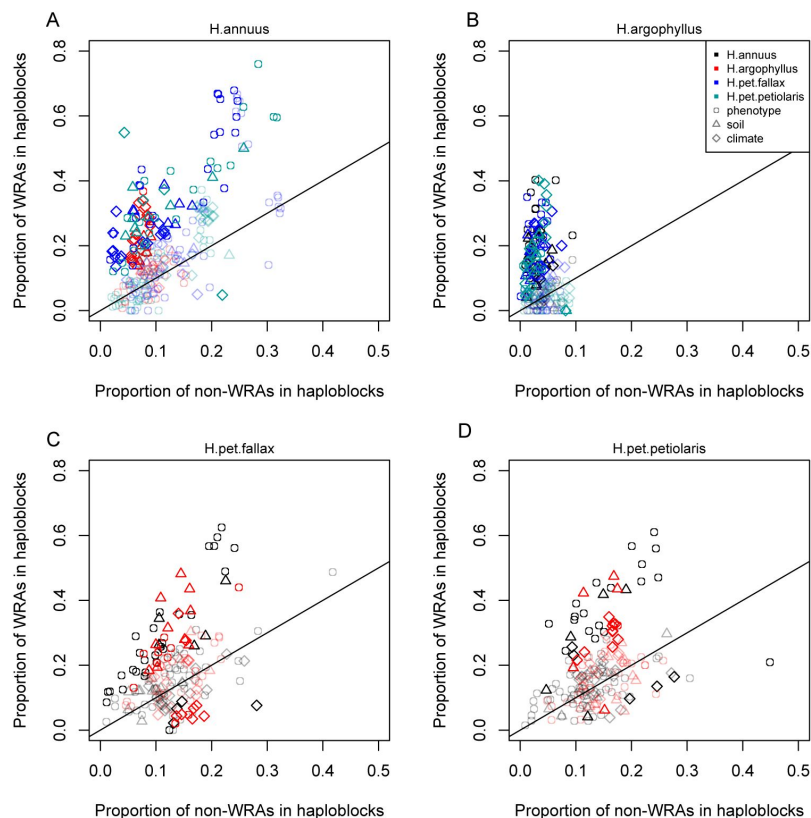


Figure 5.

Enrichment of signatures of repeated association within genomic regions harbouring a haploblock in one of the two compared lineages. Each panel shows the proportion of top candidate genes that fall within haploblocks for windows with significant signatures of repeated association by the null-W test (WRAs) vs. those with non-significant signatures (non-WRAs). Comparisons of *H. petiolaris petiolaris* vs. *H. petiolaris fallax* are omitted as they share segregating haploblocks. Each point corresponds to the results for a single phenotype or environment, with dark shading used for cases where the deviation from random for the contingency table is significant by a permutation test ($p < 0.05$), and lighter shading indicating a non-significant result.

Multi-species signatures of repeated association

As a complement to the pair-wise analyses described above, we also conducted an analysis using PicMin, which simultaneously considers any number of lineages to identify genes with particularly strong and repeated signatures of association (Booker et al. 2023a). After clustering together windows < 1 Mbp apart, we found a total of 145 regions that were significant for at least one variable or phenotype (at FDR < 0.1 applied within each variable). The largest number of significant regions for environment was found with the Number of Frost-Free Days variable (20 with FDR < 0.1 ; 44 with FDR < 0.2) and for phenotypes with Total Leaf Number, a measure of developmental timing of flowering (9 with FDR < 0.1 ; 19 with FDR < 0.2). Considering the 720 individual 5000 bp windows with FDR < 0.2 for at least one variable, 387 are within 500 bp of a genic region, representing a 2.3x enrichment relative to the rate for windows that were not significant hits using PicMin (23.3% of other windows fall within 500 bp of a genic region; X^2 test $p < 10^{-15}$).

Estimating the number of potentially adaptive loci

Even for the environment with the highest repeatability (Number of Frost-Free Days; NFFD), most of the strongest signals of association are found in only one lineage (Figure 3B). Unfortunately, while the null-W test provides strong support and controls false positives when identifying repeated association, regions of the genome with strong associations in only a single species may include a large (and unknown) number of false positives. Thus, while it is biologically interesting to know how much adaptation is non-repeated among species, it is difficult to quantify with certainty (Booker et al. 2023a; Booker et al. 2023b). Under the assumption that at least some of the non-repeated signatures of association are true positives, this implies that there is considerable genotypic redundancy, with many different ways for these species to adaptively respond to variation in the same environment. To estimate the number of windows that could potentially contribute to local adaptation for each variable (L_{eff}), we modified the method of Yeaman et al. (2018) to partially account for the effect of linkage among nearby windows (see methods). This method assumes that the windows with signatures of association in each species are a random draw from a larger number of potentially contributing windows (L_{eff}), which can be inferred based on the ratio of shared vs. non-shared windows with strong association signatures (see Methods). We find that estimates of L_{eff} tend to be large, always well over 1000 windows regardless of the trait or environment (Figure S13A). There are numerous sources of error that affect the estimation of L_{eff} , which will tend to be overestimated due to linkage (Figure S13A) or when many signatures of association are false positives (see supplementary results), and will be underestimated when some repeatability is due to shared standing variation (Yeaman et al. 2018). Even after controls for linkage and assuming a high false positive rate of 80%, estimates of L_{eff} remain in the hundreds even for the variable with the lowest value of L_{eff} (Eref; Figure S13C, D).

Discussion

Despite its critical importance in shaping the architecture of adaptation, little is known about the extent of genotypic redundancy underlying different traits (Barghi et al. 2020; Láruson et al. 2020; Yeaman 2022). Here, we have shown evidence of significant repeatability in the basis of local adaptation (Figure 4, 5), but also an abundance of species-specific, non-repeated signatures (Figure 3; S13). In particular, we find that regions of the genome that harbour inversions in one species also tend to be strongly enriched for signatures of association in other species lacking the inversion (Figure 5). Taken together, this suggests that local adaptation in these species is highly flexible -- different species apparently use quite different sets of loci to adapt to the same environment -- yet still involves some component that has minimal redundancy, with inversions playing a particularly important role. Some of the “usual suspects” show up in the set of significantly repeated loci identified by PicMin: in addition to the homologs of the *FT* (*FLOWERING LOCUS T*) gene reported by (Todesco et al. 2020), we also found hits for several other genes involved in circadian regulation and flowering time, including *PRR3* (Para et al. 2007), *TOE1* (Aukerman and Sakai 2003), and *PHYC* (Takano et al. 2005; Chen et al. 2014), which is known to regulate photoperiodic responses in *Arabidopsis* accessions (Balasubramanian et al. 2006). Other top hits included genes involved in plant development and auxin transport (*PIN3*; (Keuskamp et al. 2010), *ARF4* (Pekker et al. 2005), and *MN*; (Bhatia et al. 2016)), and plant immunity (*CRK13* (Acharya et al. 2007) and *PRR2* (Cheval et al. 2017))

The increased repeatability found in regions of the genome that harbour inversions in only one species is particularly interesting. Inversions are commonly associated with local

adaptation (Wellenreuther and Bernatchez 2018), likely because they reduce the rate at which recombination breaks up combinations of co-selected alleles (Kirkpatrick and Barton 2006), which perhaps facilitates contributions by alleles that would be individually too weakly selected to overcome swamping by migration (Bürger and Akerman 2011; Yeaman and Whitlock 2011; Schaal et al. 2022). Evidence here suggests these regions still tend to contribute to adaptation in the species lacking the recombination-suppressing effect of an inversion, consistent with a strong effect of selection relative to migration on at least one locus in the region. Given that under this model, inversions will be most likely to persist if they capture multiple locally adapted alleles, the observation of enrichment of signatures of repeated adaptation within inversions could also explain their long-term persistence. If there is variation across the genome in the density of loci with the potential to be involved in local adaptation, then the evolution and maintenance of locally adaptive inversions would be biased towards co-occurring with the loci most likely to be involved in local adaptation. If the potential for local adaptation is conserved amongst species, then these same regions are more likely to have high repeatability. In this way, repeatability in inversion-containing regions may reflect the preferential retention of inversions, as well as the retention of locally adaptive alleles. As an example, chromosome 15 harbours a large (72 Mbp) haploblock in *H. annuus* that is strongly associated with NFFD (Number of Frost-Free Days), and also shows some signatures of association in the other taxa, with particularly strong signatures of association on either side of the haploblock in *H. argophyllus* (Figure 3A). Interestingly, loci of repeated association identified by PicMin within this region include two genes whose homologs are known to regulate responses to cold: *COLD-RESPONSIVE PROTEIN KINASE 1, CRPK1* (Liu et al. 2017)) and *LATE ELONGATED HYPOCOTYL, LHY* (Mizoguchi et al. 2002; Dong et al. 2011). While many studies have demonstrated the importance of inversions for adaptation (Wellenreuther and Bernatchez 2018; Hager et al. 2022), to our knowledge only one other study has documented the involvement of the same loci making contributions in the absence of the recombination-suppressing effect of the inversion (Lee et al. 2017). This also highlights how comparative studies of a species lacking an inversion may help identify which genes are driving adaptation in another species with an inversion, as segregating inversions tend to have extensive LD that prevents identification of any potential targets of natural selection within them.

While our results suggest a large number of loci can potentially contribute to adaptation, implying high redundancy (Figure 3B; S13), there are several factors that complicate inference. Separating the effects of drift and selection to detect signatures of local adaptation is notoriously difficult, because population structure often covaries with features of the environment that drive adaptation (Lotterhos and Whitlock 2014; Hoban et al. 2016; DeRaad et al. 2021). As found by other analyses (Yeaman et al. 2016; DeRaad et al. 2021), when we use structure correction in our genotype-environment association tests, we find many fewer signatures of repeated association (Figure S6 vs. S10), likely due to the reduced power of Baypass to detect true positives when the environment covaries with population structure (Booker et al. 2023b). Here, we have side-stepped the issues involved in correction for population structure by instead relying on comparisons among species to identify loci with associations more extreme than expected by chance, using the null-W and PicMin tests. This assumes that most loci in the genome are not involved in local adaptation, so that the relatively small proportion that are driving adaptation can therefore be picked out due to their tendency to fall into the tail of the distribution of association statistics. While this approach should be relatively robust for identifying loci with repeated patterns of adaptation, there is no way to formally estimate significance of associations found in only a single species, many of which may be false positives due to covariation of population structure and environment. However, even if we assume that 80% of the observed non-repeated loci are false positives, we still find that estimates of the effective number of loci contributing to adaptation (L_{eff}) are in the hundreds (Figure S13C). As we are unable to detect loci of small effect due to the limited power, our estimates of L_{eff} will also be biased

downwards by excluding these potentially important drivers. Finally, it is also difficult to exclude the contribution of introgression or incomplete lineage sorting to the observed signatures repeated association. If a locus tended to be highly introgressed between two species in a restricted region of their range, it is possible it could also covary with environment and therefore result in a signature of repeated association that would be mistakenly interpreted as adaptation. While it is difficult to preclude this from our analysis, regions of the genome with signatures of repeated association do not tend to have higher levels of shared standing variation than background regions (Figure S5), suggesting this is not a broad explanation for our observations. If anything, shared standing variation would be expected to increase repeatability of adaptation (MacPherson and Nuismer 2017), adding further weight to the inference that there is high genotypic redundancy in these species.

In general, we find that temperature is the strongest driver of repeated adaptation at both the phenotypic and genomic levels. We quantified local adaptation at the phenotypic level using correlations between traits measured in common gardens and the home environment of the population they were sampled from. Across all phenotypes, these correlations tended to be strongest and most similar among species for temperature variables, particularly in comparisons among *H. annuus* and *H. petiolaris* subspecies (Figure S2). We see similar patterns of repeatability reflected in the genome, where temperature variables also tend to have the greatest repeatability (Figure 4). The similarity in these phenotypic and genomic signatures is consistent with an effect of strong selection, as other artefactual or drift-based explanations for repeatability would not be expected to reflect patterns found at the phenotypic level. It should be noted that the reduced importance of soil variables in the SIPEC index might be partly driven by the fact that all traits were measured on above-ground features, due to the difficulty of getting non-disruptive phenotypic measurements for roots.

Taken together, these results suggest that some fraction of the genome contributes to adaptation with low redundancy and high repeatability (which tend to be enriched within inversions), while the remainder of the adaptive response is driven by loci with high redundancy and species-specific contributions. Models in evolutionary genetics tend to focus on extremes: population genetic approaches explore cases where strong selection deterministically drives a change in allele frequency, whereas quantitative genetic approaches make the infinitesimal assumption that phenotypic change can be realized through small frequency changes at many loci (Barton et al. 2017; Barghi et al. 2020; Yeaman 2022). Our results suggest that adaptation is a complex process that does not map cleanly onto the assumptions of either approach, but that at least some component of trait variation experiences sufficiently strong selection to overcome migration and drift and behave in a way approximately consistent with population genetic models. Given the difficulty of rigorously quantifying the contribution of alleles of small effect, it remains an open question what proportion of adaptive response is governed by loci with dynamical behaviour that follow the infinitesimal assumption. This could perhaps be quantified explicitly by decomposing trait heritabilities as has been done for standing variation in humans (Visscher et al. 2017), but this would require a considerable increase in sample size as methods such as GCTA (Yang et al. 2011) do not yield accurate estimates at the sample sizes used here.

Materials and Methods

Data collection, common garden, and phenotyping

Seed samples were collected from 151 wild sunflower populations covering most of their native distributions during the summer of 2015 (*H. annuus*: 61 populations for GWAS and 71 populations for GEA, *H. petiolaris fallax*: 23 populations, *H. petiolaris petiolaris*: 18 populations and *H. argophyllus*: 30 populations, Figure 1 and Supplementary Table 1). Seeds from ten additional populations of *H. annuus* for the GEA analysis had been previously collected in the summer of 2011. Sample seeds were obtained from randomly chosen mothers, and were first germinated in a greenhouse for two weeks, later moved to an open-sided greenhouse for acclimation. Phenotypic data were collected throughout the growing season, as detailed in Supplementary Table 1. Extensive records of developmental and morphological traits throughout the growth of the plants including leaves, stem, and seeds which were collected and digitally imaged to extract relevant phenotypic data.

Similarity in Phenotype-Environment Correlation (SIPEC).

Locally adapted traits tend to exhibit strong correlations between environment and common-garden phenotype, so environments that are important to driving local adaptation should exhibit strong correlations with many traits. As an index to represent the similarity among pairs of species in an environment's average correlation across all measured phenotypes, we calculate $SIPEC = \sum_i (|r_{i,1}| + |r_{i,2}|) |r_{i,1}| |r_{i,2}| / n_p$ where $r_{i,1}$ and $r_{i,2}$ are the Pearson correlations between the environment and the i^{th} phenotype in the first and second species, respectively, and n_p is the number of phenotypes. This SIPEC index is maximized when the correlation is large in both species regardless of the direction, so it does not differentiate phenotypically convergent vs. divergent patterns of local adaptation, and provides a means of estimating the relative importance of an environment driving local adaptation across all measured phenotypes. To account for non-independence among traits, for each pair of species we fit a PCA to all measured phenotypes and use the Principal Components that collectively explain 95% of the variance, and calculate SIPEC on the correlations of these variables with environment.

Tests of SNP association with environment (GEA) and phenotype (GWAS).

A total of 39 environmental variables (21 climatic variables, 3 geographic variables, 15 soil variables Supplementary Table 1) were used for the genotype-environment association analysis (GEA). We refer to the climatic, soil and geographic variables collectively as the “environmental variables”, for simplicity. Soil data were collected by taking three to five soil samples collected at each population from across the area in which seeds were collected and submitted to Midwest Laboratories Inc. (Omaha, NE, USA) for analysis. Climate data for each population were collected over a 30-year period (1961-1990) from geographic coordinates of the locations where the samples were collected, using the software package Climate NA (Wang et al. 2012). We used the package BAYPASS version 2.1 (Gautier 2015), which provides a re-implementation of the Bayesian hierarchical model, and explicitly accounts for the covariance structure among the population allele frequencies that originate from the shared history of the populations through the estimation of the population covariance matrix. This renders the identification of SNPs subjected to selection less sensitive to the confounding effect of population demography (Günther and Coop 2013). Population structure was estimated by choosing a random and unlinked set of 10,000 SNPs and running BAYPASS under the core model (*i.e.* no covariates). Then the Bayes factors (BF) were calculated running BAYPASS under the STD covariate model to evaluate association of SNPs with environmental variables (*i.e.* adjusting for population structure). For each SNP, the Bayes

factor (denoted BF_{is} as in [Gautier 2015](#)) was presented in decibian units (db) via the transformation $10 \log_{10}(BF)$. BF_{is} relies on the importance sampling algorithm proposed by [Coop et al., 2010](#) and uses MCMC samples obtained under the core model. To produce a narrower set of outlier loci, we followed the popular Jeffreys' rule ([Jeffreys 1961](#)) that identified outlier loci with $BF \geq 10$. As genome scan methods that correct for population structure can remove some potential signals of local adaption when there is covariation between the demographic history of the species and the environmental variables or phenotypic traits of interest, we also calculated Spearman's rank correlation (ρ , uncorrected GEA) between population allele frequencies for each SNP and each environment variable. GWAS analysis was performed 86, 30 and 69 developmental and morphologic traits in *H. annuus*, *H. argophyllus* and *H. petiolaris*, respectively (Supplementary table 1). 29 variables were measured in all three focal species, and 39 were measured only in *H. annuus* and in *H. petiolaris* subspecies (Supplementary table 1). Seed and flower traits could not be collected for *H. argophyllus*, since most plants of this species flowered very late in our common garden, and failed to form fully developed inflorescences and set seeds before temperatures became too low for their survival. Population structure was controlled for in GWAS by including the first three principal components as covariates, as well as an identity-by-state (IBS) kinship matrix calculated by EMMAX ([Kang et al. 2010](#)). We ran each trait GWAS using EMMAX (v07Mar2010), as well as the EMMAX module in EasyGWAS ([Grimm et al. 2017](#)). For every SNP/peak above the Bonferroni significance threshold, candidate genes were selected within a 100 Kb interval centered in the SNP with the lowest p-value, or within the boundaries of the GWAS peak, whichever was larger. All variants used for association were initially filtered for VQSR 90% tranche, and then further filtered to only include bi-allelic SNPs genotyped in $\geq 90\%$ of samples and with a minor allele frequency $\geq 3\%$.

Identification of top-candidate windows

We calculated the bottom 0.05 quantiles for the p -values from association tests, Spearman's rank correlation (uncorrected GEA) and GWAS (corrected and uncorrected), yielding two 5% cutoffs. For each environmental and phenotypic variable, we identified all outlier SNPs as those that fell below the respective 5% cutoff. For BayPass, we considered SNPs with Bayes factor ≥ 10 as outlier SNPs. For each 5 kb window that we defined across the whole genome, we counted the number of outlier SNPs (a) and the total number of SNPs (n). To identify top-candidate windows for each variable, we compared the number of outlier SNPs per each 5 kb window to the 0.9999 quantile of the binomial expectation where the expected frequency of outlier SNPs per window is calculated as: $\rho = \sum a_i/n_i$ (summation over all 5 kb windows), calculating ρ independently for each environmental and phenotype variable and excluding windows with 0 outliers from the calculation of ρ (as per [Yeaman et al. 2016](#)). We also calculated a top candidate index using the same approach to categorize outliers, obtaining a p -value for a binomial test for the number of SNPs per window given an expected proportion of outliers (ρ ; this p -value is not exact due to non-independence of SNPs so we refer to this as an index).

Identifying outlier SNPs detected by genome scans from genome-wide distribution without accounting for local recombination rate variation can promote false positive signals in recombination cold spots (i.e., low recombination regions) and be overly conservative in recombination hot spots (i.e., high recombination regions) ([Booker et al. 2020](#)). Therefore, to account for local recombination rate variation, all genomic windows were binned by their estimated recombination rates into 5 equally sized bins (bin1: 0-20% quantile, bin2: 20%-40% quantile, bin3: 40%-60% quantile, bin4: 60%-80% quantile, bin5: 80%-100% quantile). For each recombination bin, we estimated expected frequency of SNPs per window (ρ) and calculated cut-off separately. Windows falling above the threshold were identified as top candidate windows.

Genome-wide survey of repeatability (null-W test).

To explore repeatable genomic signatures of adaptation for each of the six pairwise contrasts among the 4 taxa (Figure 1), we used the method developed by Yeaman et al. (2016) with some modifications. A common approach is to identify candidates for adaptation independently in each species and then examine the overlap between these lists, however this approach is quite stringent and may miss many interesting signals. The null-W test is more sensitive, as it takes the list of top candidates from one species and tests whether they tend to show more extreme signatures of association than expected by chance. The null-W test is especially favorable when LD increased divergence of SNPs in tight linkage with casual SNPs but did not raise the test values enough for a window to be classified as an outlier according to the binomial test. For each top candidate window that we identified for each focal species in a pair, we refer to the same window in the other species as “top candidate ortholog”. The null distribution for each focal species and variable was constructed by randomly sampling 10,000 background SNPs from non-top candidate ortholog windows. For each non-top-candidate window, we then estimated the test statistic (W) for the Wilcoxon-signed rank test vs. the 10,000 background SNPs. This resulted in a null distribution representing the differences between the 10,000 background SNPs and the non-top-candidate ortholog windows. These were then standardized to Z-scores using the method in Whitlock and Schluter (2020):

$$Z = \frac{2W - n_1n_2}{\sqrt{n_1n_2(n_1+n_2+1)/3}} \quad (\text{Equation 2})$$

where n_1 and n_2 are the sample sizes being compared. In order to control for heterogeneity in recombination rate and its possible effects on the null distribution, we estimated null distribution for each recombination bin separately (5 in total, see above). We then compared the p-values and bayes factors (BFs) for each focal top-candidate widow to the 10,000 background SNPs, calculating the W statistics and converting into a Z-score. Empirical P-values were then calculated by comparing the Z-score for each top-candidate window to the null distribution. When individual windows had values of W that exceeded the bounds of the null distribution, their empirical p -value was set to the reciprocal of the number of genes in the null distribution. For each species pair, we refer to the windows identified as significant by this test as “Windows of Repeated Association” or WRAs.

Linkage disequilibrium and detection of clusters of repeated association

Linkage disequilibrium among adjacent genomic windows can result in statistical non-independence and similar GEA/GWAS signatures across many windows. To identify the most significant WRAs and group neighbouring windows with similar signatures of repeatability into a single “Cluster of Repeated Association” (CRA), we used the following approach in each pairwise contrast, and for each environmental variable and phenotype: for each variable, empirical p -values for all WRAs were converted to q -values to adjust for False Discovery, then beginning with the first significant WRA along the chromosome (i.e. with $q < 0.05$), we compared it to the next closest significant WRA by calculating the squared Pearson correlation coefficient (r^2) on the allele frequencies across all pairs of SNPs, and compared this estimate to a null distribution. To construct the null distribution, we generated a distribution of LD measurements between 10,000 randomly chosen windows with the same physical distance as between two significant WRAs (excluding all WRAs from the null distribution). If the r^2 between the two neighboring significant WRAs was greater than the

95th percentile of the null distribution, we clustered these two windows together. This process was repeated successively with the next closest neighboring significant WRA, walking out along the chromosome until one of two stopping criteria was reached: 1) the LD between the last two windows did not exceed the 95 percent of the tail distribution or 2) the distance between the initial window and the current window next to it was larger than 1 cM, based on the linkage map from [Todesco et al. \(2020\)](#). When the first round of clustering stopped due to either of these two criteria, all the clustered windows were removed from the dataset and the process started with the second smallest empirical p-value. By doing this way, each significant WRA will only appear in a single CRA.

Estimating an index of shared standing variation

As genotype calling was conducted separately for each species (due to computational concerns), we estimated the amount of shared standing variation based on counts of shared vs. non-shared SNPs. If two species are evolving independently, the number of shared SNPs should follow a hypergeometric distribution, so we used an approach similar to the C-scores ([Yeaman et al. 2018](#)) to calculate the difference between the observed number of shared SNPs and the expectation, scaled by the standard deviation of the hypergeometric. Because of noise and a relatively small number of SNPs per window, we applied this approach on a sliding window basis, including the 5 flanking windows upstream and downstream of each focal window the calculation of its index of shared standing variation.

Correspondence between regions of repeated association and chromosomal rearrangements

To assess the extent of overlap between regions of the genome with repeated signatures of association and previously identified low-recombination haploblocks, we used two approaches. Firstly, for each pairwise species contrast and variable, we constructed a contingency table for the number of top candidates that were significant WRAs vs. not-significant WRAs (by the null-W test), and that did vs. did not fall within a haploblock. We calculated the Pearson's X^2 statistic on this table, and then permuted the location of haploblocks throughout the genome to construct a null distribution of X^2 statistics, and calculated the p-value as the proportion of the null that exceeded the observed X^2 statistic (to account for non-independence of nearby WRAs). Secondly, we compared observations of the length and number of overlapping regions to expectations based on a randomization approach. For each pair of species, we kept the position of each CRA constant and randomized the position of haploblocks 10,000 times to build a null distribution. By keeping the position of the CRAs constant, we maintained the architecture of adaptation independent from chromosomal rearrangements. We assessed significance by testing whether the observed overlaps between CRAs and haploblocks were more extreme than the 95th percentile of the tail of null distribution. Details about identifying chromosomal rearrangements can be found in [Todesco et al. \(2020\)](#).

Identifying repeated signatures of association across all taxa.

As a complement to the pairwise analysis, we used PicMin ([Booker et al. 2022a](#)) to identify windows of the genome with strong signatures of association in multiple (sub-) species. For each environmental variable, association signatures for each window are ranked genome-wide and PicMin identifies windows with extreme ranks in multiple species. We ran the analysis once with each of the two 3-way comparisons involving one *petiolaris* subspecies (i.e. *H. annuus*, *H. argophyllus*, and either *H. pet. petiolaris* or *H. pet. fallax*) to control for repeatability arising from similar patterns in the two *petiolaris* subspecies (due to high introgression/shared standing variation).

Evaluating genotypic redundancy using C-scores

Genome-wide quantification of the repeatability of association statistics provides insights into the amount of genotypic redundancy underlying a trait or environmental adaptation, and can be assessed using the C-score approach (Yeaman et al. 2018). Briefly, for a given trait or environment, the set of association test scores within each species is classified into “associated” or “non-associated” using the binomial top candidate approach. For a given pair of species ($i = 1, 2$), the observed number of associated windows in each species (a_1, a_2) can be compared to the number of windows that are associated in both species (a_b) and the total number of windows being analysed (a_x). Under a null hypothesis where all windows in the genome have equal potential to be involved in adaptation (i.e. associated with the trait or environment), the expected number of windows associated in both species will be described by a hypergeometric distribution, with the expectation $\overline{a_b} = a_1 a_2 / a_x$. The difference between the observed and expected amount of overlap in association scores can be quantified as a C-score by scaling the difference by the standard deviation of the hypergeometric (i.e. a C-score of 2 means that the observed amount of overlap is two standard deviations above the expectation under randomness).

We assess the C-scores for each phenotype and environment trait by classifying the top 0.5% of all 5 kb windows within each species as “associated” (a_i ; based on the binomial top candidate index), and begin by calculating the C-score obtained when a_x is set to the union of a_i across all four focal species/sub-species (i.e. only those windows associated in at least one species are included in a_x , a number much lower than the total number of windows in the genome). This yields a negative C-score, as a random draw from such a small number of windows tends to yield many more overlapping associations by chance than the observation. When the C-score = 0, it means that the observed overlap between the pair of taxa being considered is consistent with a random draw of their respective “associated” windows from an overall pool of a_x windows. Thus, by adding rows to the matrix with “non-associated” scores for all 4 species/sub-species until finding the matrix that yields C-score = 0, we can estimate the effective number of loci that contributed to variation for the trait or environment being considered (L_{eff}), under the assumption that all such $a_x = L_{eff}$ windows had an equal chance of contributing to the associations.

To run the C-score analysis on LD-clustered windows, we ran the following algorithm for each trait/environmental variable: for each species, we identified all CRAs that also had a top candidate index in top 0.5%. In most cases, a large cluster with associated windows in one species corresponds to either no clusters or a small cluster in another species. To harmonize cluster boundaries across species, we bin any overlapping clusters together to their maximum extent, and take the average top candidate index for each cluster in each species as either: (a) the mean across all windows that were “associated” in that species or (b) the average across all windows associated in any species, if no windows were associated in that species. This yields a matrix that can be submitted to the hypergeometric C-score analysis.

Data availability

All scripts used for analysis and figures will be deposited in a Dryad archive along data files and results. Raw sequence data are deposited in the Sequence Read Archive (SRA) under BioProject accessions PRJNA532579, PRJNA398560, and PRJNA564337, as described in Todesco et al. (2020). During review, the materials for the Dryad archive are available here: <https://www.dropbox.com/sh/cwa34fu2o7tn6dm/AADK56BnmX8IhQxVK70Evbgra?dl=0>

Acknowledgements

The authors would like to thank the Yeaman and Peichel labs for comments. Computational support was provided by the Digital Resources Alliance of Canada, and funding was provided by Genome Canada and Genome BC (LSARP2014-223SUN) and the International Consortium for Sunflower Genomic Resources. SY was funded by Alberta Innovates and NSERC Discovery. GLO was funded by a Banding Postdoctoral Fellowship.

References

- Acharya BR , Raina S , Maqbool SB , Jagadeeswaran G , Mosher SL , Appel HM , Schultz JC , Klessig DF , Raina R (2007) **Overexpression of CRK13, an Arabidopsis cysteine-rich receptor-like kinase, results in enhanced resistance to Pseudomonas syringae: Overexpression of CRK13 induces pathogen defense** *The Plant Journal* **50**:488–499
- Auge GA , Penfield S , Donohue K (2019) **Pleiotropy in developmental regulation by flowering-pathway genes: is it an evolutionary constraint?** *New Phytol* **224**:55–70
- Aukerman MJ , Sakai H (2003) **Regulation of Flowering Time and Floral Organ Identity by a MicroRNA and Its APETALA2-Like Target Genes** *Plant Cell* **15**:2730–2741
- Balasubramanian S , Sureshkumar S , Agrawal M , Michael TP , Wessinger C , Maloof JN , Clark R , Warthmann N , Chory J , Weigel D (2006) **The PHYTOCHROME C photoreceptor gene mediates natural variation in flowering and growth responses of Arabidopsis thaliana** *Nat Genet* **38**:711–715
- Barghi N , Hermisson J , Schlötterer C (2020) **Polygenic adaptation: a unifying framework to understand positive selection** *Nat Rev Genet* **21**:769–781
- Barton NH , Etheridge AM , Véber A (2017) **The infinitesimal model: Definition, derivation, and implications** *Theoretical Population Biology* **118**:50–73
- Bhatia N , Bozorg B , Larsson A , Ohno C , Jönsson H , Heisler MG (2016) **Auxin Acts through MONOPTEROS to Regulate Plant Cell Polarity and Pattern Phyllotaxis** *Current Biology* **26**:3202–3208
- Blackman BK , Scascitelli M , Kane NC , Luton HH , Rasmussen DA , Bye RA , Lentz DL , Rieseberg LH (2011) **Sunflower domestication alleles support single domestication center in eastern North America** *Proc. Natl. Acad. Sci. U.S.A* **108**:14360–14365
- Bohutínská M , Vlček J , Yair S , Laenen B , Konečná V , Fracassetti M , Slotte T , Kolář F (2021) **Genomic basis of parallel adaptation varies with divergence in Arabidopsis and its relatives** *Proc. Natl. Acad. Sci. U.S.A* **118**
- Booker TR , Yeaman S , Whitlock MC (2020) **Variation in recombination rate affects detection of outliers in genome scans under neutrality** *Mol Ecol* **29**:4274–4279

Booker TR , Yeaman S , Whitlock MC (2023) **Using genome scans to identify genes used repeatedly for adaptation** *Evolution* **77**:801–811

Booker TR , Yeaman S , Whitlock MC. (2023) **Mol Ecol Res**

Bürger R , Akerman A (2011) **The effects of linkage and gene flow on local adaptation: A two-locus continent-island model** *Theoretical Population Biology* **80**:272–288

Chen A , Li C , Hu W , Lau MY , Lin H , Rockwell NC , Martin SS , Jernstedt JA , Lagarias JC , Dubcovsky J (2014) **PHYTOCHROME C plays a major role in the acceleration of wheat flowering under long-day photoperiod** *Proc. Natl. Acad. Sci. U.S.A* **111**:10037–10044

Cheval C , Perez M , Leba LJ , Ranty B , Perochon A , Reichelt M , Mithöfer A , Robe E , Mazars C , Galaud JP , et al. (2017) **PRR2, a pseudo-response regulator, promotes salicylic acid and camalexin accumulation during plant immunity** *Sci Rep* **7**

DeRaad DA , Cobos ME , Alkishe A , Ashraf U , Ahadji-Dabla KM , Nuñez-Penichet C , Peterson AT (2021) **Genome-environment association methods comparison supports omnigenic adaptation to ecological niche in malaria vector mosquitoes** *Molecular Ecology* **30**:6468–6485

Dong MA , Farré EM , Thomashow MF (2011) **CIRCADIAN CLOCK-ASSOCIATED 1 and LATE ELONGATED HYPOCOTYL regulate expression of the C-REPEAT BINDING FACTOR (CBF) pathway in Arabidopsis** *Proc. Natl. Acad. Sci. U.S.A* **108**:7241–7246

Gautier M (2015) **Genome-Wide Scan for Adaptive Divergence and Association with Population-Specific Covariates** *Genetics* **201**:1555–1579

Grimm DG , Roqueiro D , Salomé PA , Kleeberger S , Greshake B , Zhu W , Liu C , Lippert C , Stegle O , Schölkopf B , et al. (2017) **easyGWAS: A Cloud-Based Platform for Comparing the Results of Genome-Wide Association Studies** *Plant Cell* **29**:5–19

Günther T , Coop G (2013) **Robust Identification of Local Adaptation from Allele Frequencies** *Genetics* **195**:205–220

Hager ER , Harringmeyer OS , Wooldridge TB , Theingi S , Gable JT , McFadden S , Neugeboren B , Turner KM , Jensen JD , Hoekstra HE (2022) **A chromosomal inversion contributes to divergence in multiple traits between deer mouse ecotypes** *Science* **377**:399–405

Heiser C , Smith D , Clevenger S , Martin W (1969) **The North American sunflowers (Helianthus)** *Memoirs of the Torrey Botanical Club* **22**:1–218

Heiser CB (1951) **Hybridization in the Annual Sunflowers: Helianthus annuus x H. argophyllus** *The American Naturalist* **85**:65–72

Hoban S , Kelley JL , Lotterhos KE , Antolin MF , Bradburd G , Lowry DB , Poss ML , Reed LK , Storfer A , Whitlock MC (2016) **Finding the Genomic Basis of Local Adaptation: Pitfalls, Practical Solutions, and Future Directions** *The American Naturalist* **188**:379–397

Izawa T (2007) **Adaptation of flowering-time by natural and artificial selection in *Arabidopsis* and rice** *Journal of Experimental Botany* **58**:3091–3097

Jeffreys H (1961) **Theory of probability**

Kang HM , Sul JH , Service SK , Zaitlen NA , Kong S , Freimer NB , Sabatti C , Eskin E (2010) **Variance component model to account for sample structure in genome-wide association studies** *Nat Genet* **42**:348–354

Keuskamp DH , Pollmann S , Voeselek LACJ , Peeters AJM , Pierik R (2010) **Auxin transport through PIN-FORMED 3 (PIN3) controls shade avoidance and fitness during competition** *Proc. Natl. Acad. Sci. U.S.A* **107**:22740–22744

Kirkpatrick M , Barton N (2006) **Chromosome Inversions, Local Adaptation and Speciation** *Genetics* **173**:419–434

Láruson ÁJ , Yeaman S , Lotterhos KE (2020) **The Importance of Genetic Redundancy in Evolution** *Trends in Ecology & Evolution* **35**:809–822

Lee C-R , Wang B , Mojica JP , Mandáková T , Prasad KVSK , Goicoechea JL , Perera N , Hellsten U , Hundley HN , Johnson J , et al. (2017) **Young inversion with multiple linked QTLs under selection in a hybrid zone** *Nat Ecol Evol* **1**

Liu Z , Jia Y , Ding Y , Shi Y , Li Z , Guo Y , Gong Z , Yang S (2017) **Plasma Membrane CRPK1-Mediated Phosphorylation of 14-3-3 Proteins Induces Their Nuclear Import to Fine-Tune CBF Signaling during Cold Response** *Molecular Cell* **66**:117–128

Lotterhos KE , Whitlock MC (2014) **Evaluation of demographic history and neutral parameterization on the performance of F ST outlier tests** *Mol Ecol* **23**:2178–2192

MacPherson A , Nuismer SL (2017) **The probability of parallel genetic evolution from standing genetic variation** *J. Evol. Biol* **30**:326–337

Manceau M , Domingues VS , Linnen CR , Rosenblum EB , Hoekstra HE (2010) **Convergence in pigmentation at multiple levels: mutations, genes and function** *Phil. Trans. R. Soc. B* **365**:2439–2450

Mizoguchi T , Wheatley K , Hanzawa Y , Wright L , Mizoguchi M , Song HR , Carré IA , Coupland G (2002) **LHY and CCA1 are partially redundant genes required to maintain circadian rhythms in *Arabidopsis*** *Dev Cell* **2**:629–641

Moyers BT , Rieseberg LH (2016) **Remarkable life history polymorphism may be evolving under divergent selection in the silverleaf sunflower** *Mol Ecol* **25**:3817–3830

Ostevik KL , Andrew RL , Otto SP , Rieseberg LH (2016) **Multiple reproductive barriers separate recently diverged sunflower ecotypes: reproductive barriers between sunflower ecotypes** *Evolution* **70**:2322–2335

Para A , Farré EM , Imaizumi T , Pruneda-Paz JL , Harmon FG , Kay SA (2007) **PRR3 Is a vascular regulator of TOC1 stability in the Arabidopsis circadian clock** *Plant Cell* **19**:3462–3473

Pekker I , Alvarez JP , Eshed Y (2005) **Auxin response factors mediate Arabidopsis organ asymmetry via modulation of KANADI activity** *Plant Cell* **17**:2899–2910

Rosenblum EB , Parent CE , Brandt EE (2014) **The Molecular Basis of Phenotypic Convergence** *Annu. Rev. Ecol. Evol. Syst* **45**:203–226

Sambatti JBM , Strasburg JL , Ortiz-Barrientos D , Baack EJ , Rieseberg LH (2012) **Reconciling extremely strong barriers with high levels of gene exchange in annual sunflowers: gene flow across a strong reproductive barrier** *Evolution* **66**:1459–1473

Schaal SM , Haller BC , Lotterhos KE (2022) **Inversion invasions: when the genetic basis of local adaptation is concentrated within inversions in the face of gene flow** *Phil. Trans. R. Soc. B* **377**

Strasburg JL , Rieseberg LH (2008) **Molecular demographic history of the annual sunflowers *helianthus annuus* and *h. Petiolaris*** large effective population sizes and rates of long-term gene flow *Evolution* **62**:1936–1950

Takano M , Inagaki N , Xie X , Yuzurihara N , Hihara F , Ishizuka T , Yano M , Nishimura M , Miyao A , Hirochika H , et al. (2005) **Distinct and Cooperative Functions of Phytochromes A, B, and C in the Control of Deetiolation and Flowering in Rice** *The Plant Cell* **17**:3311–3325

Tittes S , Lorant A , McGinty S , Doebley JF , Holland JB , Sánchez-González J de J , Seetharam A , Tenaillon M , Ross-Ibarra J (2021) **Tittes S, Lorant A, McGinty S, Doebley JF, Holland JB, Sánchez-González J de J, Seetharam A, Tenaillon M, Ross-Ibarra J. 2021. Not so local: the population genetics of convergent adaptation in maize and teosinte. Evolutionary Biology Available from: <http://biorxiv.org/lookup/doi/10.1101/2021.09.09.459637>**

Todesco M , Owens GL , Bercovich N , Légaré J-S , Soudi S , Burge DO , Huang K , Ostevik KL , Drummond EBM , Imerovski I , et al. (2020) **Massive haplotypes underlie ecotypic differentiation in sunflowers** *Nature* **584**:602–607

Visscher PM , Wray NR , Zhang Q , Sklar P , McCarthy MI , Brown MA , Yang J (2017) **10 Years of GWAS Discovery: Biology Function, and Translation. The American Journal of Human Genetics** **101**:5–22

Wang L , Josepchs EB , Lee KM , Roberts LM , Rellán-Álvarez R , Ross-Ibarra J , Hufford MB. (2021) **Molecular Parallelism Underlies Convergent Highland Adaptation of Maize Landraces** *Molecular Biology and Evolution* **38**:3567–3580

Wang T , Hamann A , Spittlehouse DL , Murdock TQ (2012) **ClimateWNA—High-Resolution Spatial Climate Data for Western North America** *Journal Of Applied Meteorology And Climatology* **51**

Wellenreuther M , Bernatchez L (2018) **Eco-Evolutionary Genomics of Chromosomal Inversions** *Trends in Ecology & Evolution* **33**:427–440

Whitlock M , Schluter D (2020) **The analysis of biological data** *The American Journal of Human Genetics* **88**:76–82

Yeaman S (2022) **Evolution of polygenic traits under global vs local adaptation** *Genetics* **220**

Yeaman S , Gerstein AC , Hodgins KA , Whitlock MC. (2018) **Quantifying how constraints limit the diversity of viable routes to adaptation** *PLoS Genet* **14**

Yeaman S , Hodgins KA , Lotterhos KE , Suren H , Nadeau S , Degner JC , Nurkowski KA , Smets P , Wang T , Gray LK , et al. (2016) **Convergent local adaptation to climate in distantly related conifers** *Science* **353**:1431–1433

Yeaman S , Whitlock MC (2011) **The genetic architecture of adaptation under migration-selection balance: the genetic architecture of local adaptation** *Evolution* **65**:1897–1911

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Reviewer #1 (Public Review):

Soudi, Jahani et al. provide a valuable comparative study of local adaptation in four species of sunflowers and investigate the repeatability of observed genomic signals of adaptation and their link to haploblocks, known to be numerous and important in this system. The study builds on previous work in sunflowers that have investigated haploblocks in those species and on methodologies developed to look at repeated signals of local adaptations. The authors provide solid evidence of both genotype-environment associations (GEA) and genome-wide association study (GWAS), as well as phenotypic correlations with the environment, to show that part of the local adaptation signal is repeatable and significantly co-occur in regions harboring haploblocks. Results also show that part of the signal is species specific and points to high genetic redundancy. The authors rightfully point out the complexities of the adaptation process and that the truth must lie somewhere between two extreme models of evolutionary genetics, i.e. a population genetics view of large effect loci and a quantitative genetics model. The authors take great care in acknowledging and investigating the multiple biases inherent to the used methods (GEA and GWAS) and use a conservative approach to draw their conclusions. The multiplicity of analyses and their interdependence make them slightly hard to understand and the manuscript would benefit from more careful explanations of concepts and logical links throughout. This work will be of interest to evolutionary biologists and population geneticists in particular, and constitutes an additional applied example to the comparative local adaptation literature.

Some thoughts on the last paragraph of the discussion (L481-497): I think it would be fine to have some more thoughts here on the processes that could contribute to the presence/absence of inversions, maybe in an "Ideas and Speculation" subsection. To me, your results point to the fact that though inversions are often presented as important for local adaptation, they seem to be highly contingent on the context of adaptation in each species. First, repeatability results are only at the window/gene level in your results, the specific mutations are not under scrutiny. Is it possible that inversions are only necessary when sets of small effect mutations are used, opposite to a large effect mutation in other species? Additionally, in a model with epistasis, fitness effects of mutations are dependent on the genomic background and it is possible that inversions were necessary in only certain contexts, even for the same mutations, i.e. some adaptive path contingency. Finally, do you have specific demographic history knowledge in this system that maps to the observations of the presence of inversions or not? For example, have the species "using" inversions been subject to more gene flow compared to others?

Reviewer #2 (Public Review):

In this study the authors sought to understand the extent of similarity among species in intraspecific adaptation to environmental heterogeneity at the phenotypic and genetic levels. A particular focus was to evaluate if regions that were associated with adaptation within putative inversions in one species were also candidates for adaptation in another species that lacked those inversions. This study is timely for the field of evolutionary genomics, due to recent interest surrounding how inversions arise and become established in adaptation.

Major strengths

Their study system was well suited to addressing the aims, given that the different species of sunflower all had GWAS data on the same phenotypes from common garden experiments as well as landscape genomic data, and orthologous SNPs could be identified. Organizing a dataset of this magnitude is no small feat. The authors integrate many state-of-the-art statistical methods that they have developed in previous research into a framework for correlating genomic Windows of Repeated Association (WRA, also amalgamated into Clusters of Repeated Association based on LD among windows) with Similarity In Phenotype-Environment Correlation (SIPEC). The WRA/CRA methods are very useful and the authors do an excellent job at outlining the rationale for these methods.

Major weaknesses

The study results rely heavily on the SIPEC measure, but I found the values reported difficult to interpret biologically. For example, in Figure 4 there is a range of SIPEC from 0 to 0.03 for most species pairs, with some pairs only as high as ~0.01. This does not appear to be a high degree of similarity in phenotype-environment correlation. For example, given the equation on line 517 for a single phenotype, if one species has a phenotype-environment correlation of 1.0 and the other has a correlation of 0.02, I would postulate that these two species do not have similar evolutionary responses, but the equation would give a value of $(1+0.02)*1*0.02/1 = 0.02$ which is pretty typical "higher" value in Figure 4. I also question the logic behind using absolute values of the correlations for the SIPEC, because if a trait increases with an environment in one species but decreases with the environment in another species, I would not predict that the genetic basis of adaptation would be similar (as a side note, I would not question the logic behind using absolute correlations for associations with alleles, due to the arbitrary nature of signing alleles). I might be missing something here, so I look forward to reading the author's responses on these thoughts.

An additional potential problem with the analysis is that from the way the analysis is presented, it appears that the 33 environmental variables were essentially treated as independent data points (e.g. in Figure 4, Figure 5). It's not appropriate to treat the environmental variables independently because many of them are highly correlated. For example in Figure 4, many of the high similarity/CRA values tend to be categorized as temperature variables, which are likely to be highly correlated with each other. This seems like a type of pseudo replication and is a major weakness of the framework.

Below I highlight the main claims from the study and evaluate how well the results support the conclusions.

* "We find evidence of significant genome-wide repeatability in signatures of association to phenotypes and environments" (abstract)

* Given the questions above about SIPEC, I did not find this conclusion well supported with the way the data are presented in the manuscript.

* "We find evidence of significant genome-wide repeatability in signatures of association to phenotypes and environments, which are particularly enriched within regions of the genome harbouring an inversion in one species. " (Abstract) And "increased repeatability found in regions of the genome that harbour inversions" (Discussion)

* These claims are supported by the data shown in Figure 4, which shows that haploblocks are enriched for WRAs. I want to clarify a point about the wording here, as my understanding of the analysis is that the authors test if *haploblocks* are enriched with *WRAs*, not whether *WRAs* are enriched for *haploblocks*. The wording of the abstract is claiming the latter, but I think what they tested was the former. Let me know if I'm missing something here.

* Notwithstanding the concerns about highly correlated environments potentially inflating some of the patterns in the manuscript, to my knowledge this is the first attempt in the literature to try this kind of comparison, and the results does generally suggest that inversions are more likely capturing, rather than accumulating adaptive variation. However, I don't think the authors can claim that repeated signatures are enriched with haploblock regions, and the authors should take care to refrain from stating the relative importance of different regions of the genome to adaptation without an analysis.

* "While a large number of genomic regions show evidence of repeated adaptation, most of the strongest signatures of association still tend to be species-specific, indicating substantial genotypic redundancy for local adaptation in these species." (Abstract)

* Figure 3B certainly makes it look like there is very little similarity among species in the genetic basis of adaptation, which leaves the question as to how important the repeated signatures really are for adaptation if there are very few of them. (Is 3B for the whole genome or only that region?). This result seems to be at odds with the large number of CRAs and the claims about the importance of haploblock regions to adaptation, which extend from my previous point.

* "we have shown evidence of significant repeatability in the basis of local adaptation (Figure 4, 5), but also an abundance of species-specific, non-repeated signatures (Figure 3)"

* While the claim is a solid one, I am left wondering how much of these genomes show repeated vs. non-repeated signatures, how much of these genomes have haploblocks, and how much overlap there really is. Finding a way to intuitively represent these unknowns would greatly strengthen the manuscript.

Overall, I think the main claims from the study, the statistical framework, and the results could be revised to better support each other.

Although the current version of the manuscript has some potential shortcomings with regards to the statistical approaches, and the impact of this paper in its present form could be stifled because the biology tended to get lost in the statistics, these shortcomings may be addressed by the authors.

With some revisions, the framework and data could have a high impact and be of high utility to the community.